Terry Tao

portfolio

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SpaceX (Starlink) | REDMOND, WA — AUG 2026 – PRESENT
Flight Software Engineer | 5G Direct to Cell

- Wrote bespoke Rust stack for 5G radio protocol generalized for both RAN (sat side) and UE (phone side) on a team of four
- Designed modular code to swap between pure CPU and ASIC implementations using Rust traits
- Ensured deterministic behavior by standardizing a TX and RX function call chain
- Questioned a requirement to simulate link effects at the L1 boundary, saving unnecessary engineering hours
- Deleted duplicate code by storing each slot's grants and data in a ring buffer that advances every slot
- Enabled faster/slower-than-wall-time execution by standardizing a single source of truth for time based on I/Q sample time

SpaceX (Starlink) | REDMOND, WA — MAY 2025 – AUG 2025
Flight Software Intern | 5G Direct to Cell

- Implemented 5G Physical Layer channel coding (LDPC) and modulation (QAM) in Rust
- Optimized LDPC min-sum decoder with AVX512 SIMD intrinsics to get 10Mb/s throughput on one core
- Proved correctness by connecting a third-party test UE (phone) to the Rust stack over a Software Defined Radio

University of Michigan - Ann Arbor | ANN ARBOR, MI — SEPT 2022 – APRIL 2026
M.S Robotics Engineering | GPA 3.5 / 4.0
B.S.E Robotics Engineering | GPA 3.7 / 4.0

Formula SAE (MRacing) | ANN ARBOR, MI — SEPT 2022 – JUL 2026
Autonomous Director + Founder of Autonomous System

- Led a team of 20 members including 4 subsystem leads
- Wrote firmware using asynchronous embedded Rust as well as in C using STM32CUBE
- Standardized DBC packing / unpacking teamwide with a custom C / Rust source code generator written in Rust
- Retrofit MR24 with sensors, compute, power steering, emergency brakes, and remote shutdown wiring
- Designed modular PCB for quick swapping of ECU, debug probe, GMSL2 deserializer, and GNSS receiver using M.2 connectors
- Validated mapping algorithm with real driving data to achieve a 0.1m covariance between inference and ground truth
- Designed Jetson-to-vehicle interface board for power distribution, ADC, shutdown circuit relays, status indicators, and CAN over USB
- Ensured 50% braking performance under any single point failure with DFMEA and failure tree analysis after designing e-brakes
- Managed sponsorships of over $40,000 worth of camera, LiDAR, GNSS and compute hardware

H3D Gamma | ANN ARBOR, MI — MAY 2024 – AUGUST 2024
SLAM Intern

- Automated extrinsic lidar-camera alignment for H3D's Jetson environment using a targetless method
- Advised H3D to not fuse GNSS and instead solely rely on the INS after evaluating GNSS as a source of odometry within Cartographer
- Decreased Cartographer CPU runtime by 50% by building OpenBLAS from source with ARMv8 optimizations
- Simulated radiation in a Nav2 and Gazebo environment to test control policies for a future radiation mapping robot.
- Tested a centroid-sampling-based frontier exploration policy to search any indoor environment for radiation in simulation

Ford Motor Company | ALLEN PARK, MI — MAY 2023 – AUGUST 2023
ADAS L3 Self Driving Intern

- Flagged DOC (Duty of Care safety envelope) violation events during L3 test drives using a custom vehicle deceleration model
- Compared driving policies based on DOC violation frequency to find which policy had an acceptable risk for Ford
- Synchronized CAN logs with INS logs by automating data acquisition and post processing to determine root cause of flagged events

UWB Mesh Dolphin Tracker | Capstone Project — SPRING 2025

- Designed a system to trilaterate a tag using time-of-flight UWB chirps and a central LORA base station
- Responsible for the PCB design and bringup for the anchor boards and the tag boards
- Designed a radio protocol to minimize chirp overlaps and maximize ranging rate
- Tested the system in a large atrium by walking around with the tag and watching the fused output on video

SLAM Robot | Class Project — FALL 2023

- Applied a model-based state estimator, LiDAR occupancy grid mapping, particle filter for fusing, and an exploration policy
- Result is a robot capable of mapping out a small maze and finding the exit autonomously

Rust | C | C++ | Altium | Git | MATLAB | Simulink | LaTeX | Linux | Vector Informatik | CAN | RTK GNSS | Fusion 360